

Faculty of Science, Engineering and Built Environment

SIT199 Applied Algebra and Statistics

Deakin University Unit Guide

Trimester 1, 2025

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Welcome

This unit includes: the algebra of complex numbers, matrices, probability, and the major discrete and continuous probability distributions. The relationship algebraic and transcendental functions and complex numbers are emphasised and applied to electrical networks. Solutions to systems of linear equations using matrices and determinants, and fundamental concepts vectors are considered. The statistics techniques and examples studied are relevant to the sciences in general, and engineering in particular.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

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Other members of the team and how to contact them

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Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).
In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes:

- The unit content was well organised and the learning resources were really good.
- The use of Bolster, a learning environment, was very helpful as a tool for working on exercises and learning the relevant theory.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who

have undertaken this unit in previous trimesters:

- An end-of-unit assessment (written exam) was replaced by an intra-trimester assessment task.
- We continued improving the integration of Bolster into the unit.
- With the availability of short videos covering most unit topics, lectures now function as workshops, focusing on solutions and illustrations of examples rather than theory.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Model and describe real-world problems using algebra and statistics.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy
ULO2	Utilise the notions of vectors and matrices in modelling and computing quantities to interpret the outcomes of real-word phenomena.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy
ULO3	Apply tools and concepts to determine states of a system of linear equations originating from real-world applications.	GLO1: Discipline-specific knowledge and capabilities GLO3: Digital literacy
ULO4	Critically evaluate solution approaches and tools used in solving problems.	GLO4: Critical thinking
ULO5	Communicate results by showing steps of solution approaches and interpreting the outcomes.	GLO2: Communication

Assessing your achievement of the unit learning outcomes

Summative assessment (tasks that will be graded or marked)

NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.

Summative assessment task 1

Title	Quizzes (x 5)
Brief description of assessment task	In this assessment task students are asked to individually complete a series of quizzes related to topics covered during the trimester.

	It includes written answers to mathematical problems, including full explanations of how each answer is derived. Problem solving strategy, reasoning, and communication of the solution is important in this task. The intention of the assignments is to assist students in understanding the unit contents, and to provide feedback on how to prepare a mathematical solution.
Detail of student output	This is an individual assessment task. Students will be required to solve problems using appropriate procedures as required. To be successful, students are required to prepare a written mathematical solution clearly, using appropriate wording along with mathematical symbols and notation. Clear and logical explanation of how a solution is obtained is as important as the solution itself.
Grading and weighting (% total mark for unit)	15% graded. Students will be given a numerical mark.
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: through answering conceptual and real-world questions that require algebraic or statistical modelling. ULO3: through solving conceptual and real-world problems using matrix algebra and systems of linear equations. ULO4: through evaluating different solution methods for efficiency, accuracy, and applicability, justifying their choices.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: through properly using the concepts and tools learned for mathematical modelling. GLO4: through your ability to critically analyse a solution method and justify the strategy used to solve the problem.
How and when you will receive feedback on your work	Students will receive feedback consisting of a mark and written comments via the unit site, generally within ten working days after the submission deadline. In addition, they receive feedback on solving similar problems during the practicals or lectures.
When and how to submit your work	Students must submit their response to assignment questions electronically, as specified in the assignment description, to the dropbox in the unit site. 1. **Quiz 1** is due before 8pm (AEDT) Thursday 13 March 2025 - Week 2. 2. **Quiz 2** is due before 8pm (AEDT) Thursday 27 March 2025 - Week 4. 3. **Quiz 3** is due before 8pm (AEST) Thursday 10 April 2025 - Week 6. 4. **Quiz 4** is due before 8pm (AEST) Thursday 1 May 2025 - Week 8. 5. **Quiz 5** is due before 8pm (AEST) Thursday 15 May 2025 - Week 10.

Summative assessment task 2

Title	**Problem Solving Task 1**
Brief description of assessment task	In this assessment task students are asked to individually complete a series of problems related to topics covered during the first four weeks of trimester. It includes written answers to mathematical problems, including full explanations of how each answer is derived. Problem solving strategy, reasoning, and communication of the solution is important in this task. The intention of the assignment is to assist students in understanding the unit contents, and to provide feedback on how to prepare a mathematical solution.
Detail of student output	This is an individual assessment task. Students will be required to solve problems using appropriate procedures as required. To be successful, students are required to prepare a written mathematical solution clearly, using appropriate wording along with mathematical symbols and notation. Clear and logical explanation of how a solution is obtained is as important as the solution itself.

Grading and weighting (% total mark for unit)	25% graded Students will be given a numerical mark.
This task assesses your achievement of these Unit Learning Outcome(s)	ULO2: through using vectors and matrices to model and compute real-world phenomena, demonstrating their understanding of vector and matrix operations and their applications. ULO3: through formulating and solving linear equation models in engineering, economics, and physics contexts using Gaussian elimination, inverse matrices, and other methods. ULO4: through assessing and comparing different solution methods for efficiency, accuracy, and suitability, justifying their choices and responding to questions on alternative approaches. ULO5: through communicating results by showing the steps of solution approaches and interpreting the outcomes.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: through properly using the concepts and tools learned for mathematical modelling. GLO2: through your ability to communicate their workings/ strategies to solve a problem via a written report. GLO3: through your ability to verify solutions to mathematical problems using mathematical software such as sagemath. GLO4: through your ability to critically analyse a solution method and justify the strategy used to solve the problem.
How and when you will receive feedback on your work	Students will receive feedback consisting of a mark and written comments via the unit site, generally within ten working days after the submission deadline. In addition, they receive feedback on solving similar problems during the practicals or lectures.
When and how to submit your work	Students must submit their response to assignment questions electronically, as specified in the assignment description, to the dropbox in the unit site **before 8pm (AEST), Thursday 27 March 2025 - Week 4.**

Summative assessment task 3

Title	**Problem Solving Task 2**
Brief description of assessment task	In this assessment task students are asked to individually complete a series of problems related to topics covered during the first eight weeks of trimester.
	It includes written answers to mathematical problems, including full explanations of how each answer is derived. Problem solving strategy, reasoning, and communication of the solution is important in this task. The intention of the assignment is to assist students in understanding the unit contents, and to provide feedback on how to prepare a mathematical solution.
Detail of student output	This is an individual assessment task. Students will be required to solve problems using appropriate procedures. To be successful, students must prepare a written mathematical solution clearly, using appropriate wording along with mathematical symbols and notation. Clear and logical explanation of how a solution is obtained is as important as the solution itself.
Grading and weighting (% total mark for unit)	30% graded Students will be given a numerical mark.

This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: through answering conceptual and real-world questions that require algebraic or statistical modelling. ULO4: through assessing and comparing different solution methods for efficiency, accuracy, and suitability, justifying solution methods and responding to questions on alternative approaches. ULO5: through presenting clear, structured solutions in assignments.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1: through properly using the concepts and tools learned for mathematical modelling. GLO2: through your ability to communicate your workings/ strategies to solve a problem via a written report. GLO3: through your ability to verify solutions to mathematical problems using mathematical software such as sagemath. GLO4: through your ability to critically analyse a solution method and justify the strategy used to solve the problem.
How and when you will receive feedback on your work	Students will receive feedback consisting of a mark and written comments via the unit site, generally within ten working days after the submission deadline. In addition, they receive feedback on solving similar problems during the practicals or lectures.
When and how to submit your work	Students must submit their response to assignment questions electronically, as specified in the assignment description, to the dropbox in the unit site **before 8pm (AEST) Thursday 8 May 2025 - Week 9.**

Summative assessment task 4

Title	**Problem solving and presentation task**
Brief description of assessment task	In this assessment task students are asked to individually complete a series of problems related to topics covered during the trimester. It includes written answers to mathematical problems, including full explanations of how each answer is derived. Problem solving strategy, reasoning, and communication of the solution is important in this task. The intention of the assignments is to assist students in understanding the unit contents, and to provide feedback on how to prepare a mathematical solution. In addition, students are required to investigate a topic independently and produce a written/oral presentation on the results of their investigations.
Detail of student output	This is an individual assessment task. Students will be required to investigate and solve problems independently. To be successful, students must also prepare a written and oral presentation that explains the results of their investigations clearly, using appropriate wording along with mathematical symbols and notation. Clear and logical explanation of how a solution is obtained is as important as the solution itself.
Grading and weighting (% total mark for unit)	30% - graded Students will be given a numerical mark.

This task assesses your achievement of these Unit Learning Outcome(s)	This assessment requires the solution of mathematical problems covering ULO1, ULO2, ULO3, ULO4, ULO5 and previous mathematical knowledge. The mark provides students with direct evidence of their progress towards achieving each of these ULOs. ULO2: through using vectors and matrices to model and compute real-world phenomena, demonstrating their understanding of vector and matrix operations and their applications. ULO3: through formulating and solving linear equation models in engineering, economics, and physics contexts using Gaussian elimination, inverse matrices, and other methods. ULO4: through assessing and comparing different solution methods for efficiency, accuracy, and suitability, justifying solution methods and responding to questions on alternative approaches. ULO5: through presenting clear, structured solutions in assignments, engage in verbal communication of findings, and demonstrate logical reasoning and clarity.
This task assesses your achievement of these Graduate Learning Outcome(s)	This assessment contains mathematical problems which are to be solved (GLO1, GLO2, GLO3, GLO4) and is to be done independently of other students. GLO1: applying mathematical concepts and tools to model real-world problems, analysing scenarios and constructing appropriate mathematical models. GLO2: providing clear, structured explanations of their problem-solving strategies, demonstrating logical reasoning and methodological clarity in mathematical solutions. GLO3: implementing algorithms or computational techniques in mathematical software such as SageMath to verify and validate solutions. Student's apply SageMath to solve and cross-check solutions. GLO4: evaluating different solution methods for efficiency, accuracy, and applicability, justifying their choice of problem-solving strategies. Student's explain solution methods, defend their choices, and respond to alternative viewpoints.
How and when you will receive feedback on your work	Deakin University will release the final assessment results at the stipulated timeframe. Students will receive a final mark and grade for SIT199, but students must individually make an appointment with the Unit Chair to seek feedback about the assessment.
When and how to submit your work	Students must submit their response to assignment questions electronically, as specified in the assignment description, to the dropbox in the unit site before **8pm AEST by Thursday 22 May 2025 - Week 11**

Formative assessment

(for practice and feedback rather than marks and grades)
In addition to the above, you will have a better chance of success in this Unit if you complete these tasks – you will get valuable experience and feedback.

Title	**Formative diagnostic quiz**
Brief description	This task is optional. In this assessment task students are asked to individually complete a series of online questions to identify the prior knowledge areas and skills they are lacking in. Students will have the opportunity to participate in an online diagnostic quiz in the first week. Based on the results of the quiz, students will be directed to further resources to overcome the gaps in their knowledge.

This task assesses your achievement of these Unit Learning Outcome(s)	This online quiz does not contribute to any of the ULOs.
When and how to submit your work	Students will do this quiz before the end of the first week of trimester.
How and when you will receive feedback on your work	Online quiz. Students will receive immediate feedback on the completion of the quiz. This will identify the areas where there are gaps in the prior knowledge and skills.

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

3 x 1 hour online lectures per week, 1 x 1 hour practical experience (workshop) per week.
Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including 3 x 1 hour online lectures per week (recordings provided), 1 x 1 hour online practical experience (workshop) per week.
Typical study commitment

Students will on average spend 150 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.
To achieve the SIT199 learning outcomes, students should begin by noting the important dates and times.

For campus students, the lecture and workshop timetable is known in advance of the start of trimester. Please contact the Unit Chair if your timetable does not show **3 online lectures and 1 workshop.**

The assignment deadlines all appear in the relevant unit site folder. Note that other units may have assignment deadlines at about the same time as those for SIT199. Knowing each assignment deadline will enhance planning and time management for all units.

Campus students are strongly advised to attend the online lectures and workshops, even though the notes and lecture recordings are made available in the unit site. Time spent listening and taking notes in lectures and workshops reduces the time needed to review the material later.

Online students should set aside time to read the unit material, view lecture recordings and videos, and attempt the problem set exercises and assignments. All this material is available in the unit site. Work, family, and other commitments may mean, however, that available study time varies from week to week.

The quizzes cover topics encountered during the whole trimester, and each quiz covers about two weeks of the unit content. The first assignment covers about four weeks of the unit content, the second assignment covers about eight weeks of content, and the third assignment may include any of the unit content.

To meet assignment deadlines, it is recommended that students attempt questions regularly. For instance, the first one-third of the first assignment can be completed after studying the Week 1 lecture and practical class. This will make learning easier and will avoid last minute cramming.

The lecture material is related to the questions found in the workshops and the assignments.

Note that there is no textbook for SIT199. The lectures, workshops, and the learning resources on the unit site contain all the information and examples required to complete this unit.

For students who have not studied Mathematics for several years, any revision of Year 12 Maths, particularly basic algebra, will be useful. Any problems or worries can normally be eased by contacting the lecturer.
Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SIT199 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

There is no prescribed textbook for this unit.

All essential SIT199 learning resources are available in the unit site.
The lecture notes, practical workshop notes, problem sets, assignments, recorded classes, and problem-solving videos will be provided on the unit site.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

While reference books are not essential to succeed in this unit, they provide alternative views of the material. The combination of the following books will cover the unit material, and so students are encouraged to consult them.

1. Trigonometry (2019) by Katherine Yoshiwara. Freely available at https://yoshiwarabooks.org/trig/trig.html

2. Modelling, Functions, and Graphs: Algebra for College Students (2019) by Katherine Yoshiwara. Freely available at https://yoshiwarabooks.org/mfg/MFG.html

3. Linear Algebra with Applications (2019) by W. Keith Nicholson. Freely available at https://lyryx.com/linear-algebra-applications/

4. Openintro Statistics 4th Ed (2019) by David Diez, Mine Cetinkaya-Rundel, and Christopher Barr. Freely available at https://leanpub.com/openintro-statistics

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025
Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025
Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

The unit content for each week is provided in the following table.

Week	Commencing	Topic	Assessment due date
1	3 March 2025	Trigonometry	
2#	10 March 2025	Exponential and logarithmic functions	**Assessment 1: Quiz 1** due 13 March
3	17 March 2025	Complex numbers	
4	24 March 2025	Systems of linear equations	**Assessment 1: Quiz 2** due 27 March **Assessment 2: Problem Solving Task 1** due 27 March
5	31 March 2025	Matrix algebra	
6	7 April 2025	Linear transformation and determinants	**Assessment 1: Quiz 3** due 10 April
7*~	14 April 2025	Eigenvalues, eigenvectors, and diagonalisation	
8	28 April 2025	Vector geometry	**Assessment 1: Quiz 4** due 1 May
9	5 May 2025	Probability	**Assessment 3: Problem Solving Task 2** due 8 May
10	12 May 2025	Distributions of random variables	**Assessment 1: Quiz 5** due 15 May
11	19 May 2025	Revision	**Assessment 4: Problem solving and presentation task** due 22 May
12	26 May 2025		Study period
13-14^	2 June 2025		End-of-unit assessment and exam period

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed